

Structural and dynamics of apoA-1 mimetic peptide lipid nanodisc assemblies: A molecular dynamics study

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ABSTRACT

Apolipoprotein A-I (apoA-I) mimetic peptides, inspired by the principal protein component of high-density lipoprotein, are known to self-assemble with lipids to form discoidal nanodiscs. These ones are widely used in biomedical research and as versatile scaffolds for characterization of membrane proteins in structural biology [1]. Since there is a huge strong interplay between lipids and apoA-I mimetic peptides, a precise knowledge of the composition, structure and dynamics of these complexes is need at the molecular level. Here, we investigated the 14A apoA-I mimetic, quantifying its orientation around the lipid bilayer and identifying the interactions that are crucial for nanodisc stability and dynamics using all-atom molecular dynamics simulations. To assess model fidelity, we back-calculated solid-state NMR observables, namely ¹⁵N chemical shifts and ²H quadrupolar splittings from the trajectories and compared with solid-state NMR data. For DMPC lipids, simulations support a dimeric, antiparallel, belt-like arrangement of 14A peptides around the discoidal bilayer, stabilized by π - π stacking between aromatic residues and by electrostatic and hydrophobic peptide-lipid interactions. These interactions yield structurally stable nanodiscs with pronounced heterogeneity in lipid ordering and bilayer thickness between the nanodisc center and rim [2]. Collectively, our MD results provide atomistic evidence for previously hypothesized peptide-peptide and peptide-lipid interactions and clarify how amphipathic helices organize to form the rim of discoidal nanodiscs [3]. Building on this validated model, we are extending the work from single-lipid bilayers to more realistic membrane environment using all-atom molecular dynamics, with the aim of establishing how native-like membrane complexity shapes both the peptide scaffold and the surrounding bilayer. Collectively, these insights inform the rational design of apoA-I mimetics for biomedical applications and the optimization of nanodiscs as platforms for studying membrane proteins.

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